

The OFF/ON Mechanism Study

First population-level result: dopamine, the drive, and the near-Markovian medium

Working note — fracres project (D. R. Chin)

15 September 2026

Abstract

First population study on ds002778 (commits 5523670, a9e599c, 04bb8de): for every subject $\times$ session $\times$ {Fz, Cz, Pz}, the best-fit generative mechanism (α, H, λ) was grid-searched against the channel's (DFA- H , spectral- β) statistics, with the fBm drive promoted to a first-class citizen of **fracres.drivers** and the config system. Headline: the phantom class fits *all* 44 usable recordings well; the fractional order α is near-Markovian (≈ 0.92 – 0.94) and group-invariant, so the LRD lives in the inferred drive, not the medium; and the two candidate effects — PD vs. control at Cz, and medication OFF $\rightarrow$ ON whitening — are consistent in direction across model-based and model-free estimators but underpowered at this n . Two analysis failures (a provenance trap and a grid-boundary artifact) are recorded with their resolutions.

1 Design

All 31 ds002778 subjects (15 PD $\times$ ses-off/ses-on, 16 HC $\times$ ses-hc), channels Fz/Cz/Pz, recordings resampled to 64 Hz and cropped to a uniform 120 s (7,680 samples, so estimator finite-sample biases are comparable across subjects and sessions). Grid: $\alpha \in \{0.85, 0.9, 0.95, 1.0\}$, $H \in [0.15, 0.80]$ at 0.05 spacing, $\lambda = 1, 3$ seeds per point with common random numbers; objective = Euclidean distance on (DFA- H , β), model vs. data. v2 excluded two ON sessions (below) and extended the H grid downward; incremental resumable CSV (outputs/off_on_mechanisms_v2.csv, 132 rows = 44 recordings $\times$ 3 channels).

2 Results

1. The phantom class is adequate for real EEG. Objectives are 0.046–0.092 across *all* groups and channels — the same scale as the near-exact single-subject fit. An fBm-driven contractive fractional reservoir reproduces the (DFA- H , β) signature of real resting-state scalp EEG; this validates the forward-modelling premise of the whole project at population scale.

2. The medium is near-Markovian; the LRD lives in the drive. Best-fit $\alpha = 0.92$ – 0.94 with no group or session differences (± 0.05 – 0.06 spread, means indistinguishable). Best-fit H does the work: ≈ 0.52 – 0.63 depending on site and group. Within this model class, inter-subject and inter-session variability in cortical LRD is attributed to the *driving process's* Hurst exponent, not to the medium's fractional order. (Caveat: α near the grid boundary — the class may simply not reward $\alpha \neq 1$ strongly; see open threads.)

3. PD vs. control: reduced persistence at Cz (suggestive). At Cz, HC data DFA- $H = 1.41$ vs. PD-off 1.35 (Mann–Whitney $p = 0.046$); model-based best- H agrees in direction (0.631 vs. 0.577, $p = 0.078$). Fz and Pz show nothing ($p \geq 0.10$). Model-free and model-based estimators converging at the same electrode is the strongest internal evidence; but $p = 0.046$ does not survive a 3-channel Bonferroni ($\alpha^* \approx 0.017$), so this is *suggestive*, not confirmatory.

4. Medication OFF→ON: consistent whitening, underpowered. Paired across the 13 usable PD subjects: $\Delta H = -0.01 / -0.06 / -0.04$ at Fz/Cz/Pz (Wilcoxon $p = 0.89/0.12/0.18$); model-free ΔDFA at Cz = -0.05 ($p = 0.068$). Every channel points the same way — dopamine *whitens* the effective drive (reduces persistence) — but $n = 13$ paired subjects cannot resolve an effect of this size.

channel	group/session	best H (mean $\pm$ sd)	α	obj.
Fz	HC / PD-off / PD-on	0.52/0.54/0.53	≈ 0.93	0.06–0.08
Cz	HC / PD-off / PD-on	0.63/0.58/0.52	≈ 0.92	0.05–0.07
Pz	HC / PD-off / PD-on	0.53/0.54/0.50	≈ 0.92	0.05–0.09

Table 1: Best-fit mechanism by site and group (v2, extended grid).

3 Failures encountered and how they were resolved

The provenance trap (v1’s “5× worse ON fits”).

v1 showed ON-session objectives ≈ 0.35 vs. ≈ 0.07 for everything else, and ON data statistics with tripled variance. The cause was not medication: `sub-pd6` and `sub-pd16` ON sessions are reconstructed from *preprocessed EEGLAB .mat* files (flagged in `participants.tsv` notes), and the preprocessing destroys the $1/f$ structure — their ON data β is *negative* ($-0.42, -0.24$) vs. ≈ 1.6 for everyone else. Excluding them, ON fits are as good as OFF. *Resolution*: the exclusion is encoded in the study script with the reason, not applied post-hoc. *Lesson*: read the provenance column *before* running 46 recordings; and a fit residual is a data-quality detector — the model flagged the bad rows before we did.

The grid-boundary artifact (v1’s “ON regime extends below 0.45”).

In v1 ($H \geq 0.45$), 31–46% of ON best fits piled up at the grid minimum, suggesting the ON regime lived below the grid. v2 extended H down to 0.15: *zero* best fits at the new floor; the ON distribution settles in $[0.35, 0.70]$. The “downward pull” was truncation, not signal. *Lesson*: a pile-up at a grid edge is always a grid diagnostic first and a finding second; extend before interpreting.

Two test-caught subtleties in the fBm driver.

(i) `diff` $\circ$ `cumsum` is *not* bit-identical: floating-point cancellation at `cumsum` magnitude $O(70)$ leaves a $\sim 10^{-14}$ residual — assert `atol=1e-10`, not equality. (ii) z-scoring is *affine* $((x - \mu)/\sigma)$, so the increments are pure-scaled but the trace is not — the ratio check belongs on `diff`. Both were caught by tests, not inspection.

4 Validation patterns that earned trust

- **Model-based / model-free cross-validation.** Every candidate effect was computed both from the best-fit mechanism (H) and directly from the data (DFA- H , β). Both candidate

effects (Cz group difference, OFF→ON whitening) appear in both views with the same sign — agreement between an inverse-problem estimate and a direct estimator is much stronger evidence than either alone.

- **Uniform-length crops.** Cropping every recording to 120 s made estimator finite-sample bias a constant across subjects and sessions, removing a confound that the raw files (192–213 s) would have introduced.
- **Versioned grids with named artifacts.** Keeping v1’s CSV and script behaviour reproducible made the boundary artifact demonstrable rather than anecdotal.

5 Open threads

- **Confirmation cohort.** Both candidate effects need $n \gg 13$: OpenNeuro `ds008768` (312 participants, 358 resting-state PD/HC recordings) is the obvious next dataset; the ingestion module is dataset-agnostic BIDS, so this is a mirror + config change.
- **α at the boundary.** Best-fit $\alpha \approx 1$ everywhere either means the medium is Markovian or that the objective is insensitive to α near 1 at this T . A sensitivity study (profile the objective along α at fixed best H) would distinguish these.
- **Multi-channel readout / spatial model.** Channels are currently fit independently; the neural-field variant with a leadfield-like readout (Next 2) would couple them spatially and likely sharpen the Cz effect.
- **Avalanche/criticality statistics** have not yet entered the objective; adding them (Next 3 machinery) tests whether the phantom matches more than second-order structure.